Street Fighter 6 - Input Engine Details

Buffering

SF6 has a universal buffer when it comes to input buffers for:

- After normal recover (link buffer)
- After dash
- After special move recovery
- For specials
- For dashes
- On wake up (reversal buffer)

All of those have **5 possible frames** for input (1 for frame perfect input + 4 buffered inputs).

This is similar to SFV. The only thing that SFV had different was 8f for buffering dashes.

In SF4 it was more lenient on special reversal but had no link buffer.

Motion Definitions

Notation:

- arrows like is a direction obviously
- is neutral
- (6) is the forward direction pressed for 6 frames. If not indicated any amount is ok. If a range is written like 1-11 it can be between 1 and 11 (included).
- lenient means that the input can be any combination (e.g. forward can be up-forward or down-forward)

You can assume that forward / backward variations are similar.

Dash

A valid dash corresponds to:

Input	Frames	Matching
	any	strict
	1-8	strict
	1-8	strict
	any	strict

Contrary to normal motions, the matching doesn't allow any other input in between!

Example of invalid dashes:

-
-
- (9)

Down Down

A valid "down down" input is defined by:

Input	Frames	Matching
	1+	strict
	1-10	strict
	1-10	strict
	1-10	strict

This pattern is similar as SFV but buffering is stricter as it was 12 frames before (10 now).

Quarter Circle

A valid quarter circle forward is defined by:

Input	Frames	Matching
	1-11	strict
	1-11	strict
	1-11	strict

Because it uses the normal matching procedure, you can have other inputs in between like:

- (assuming timing is ok)

This pattern is similar as SFV but buffering is stricter as it was 12 frames before (11 now).

Example of an invalid QCF:

-

In some cases, we've noticed that you can input QCF by performing an HCF which has a higher priority. Example: Ryu's fireball. Currently, it only exists on motions where you have a conflict between QCF and DP. Example: Manon's Renversé (QCF P) cannot use an HCF input.

Dragon Punch Motion

The DP is defined by

Input	Frames	Matching
	1-7	lenient
	1-7	lenient
	1-7	lenient

Some examples (assuming timing is within constraints):

-
-
-

This pattern is similar as SFV but buffering is stricter as it was 8 frames before (7 now).

Double Quarter Circle

A double quarter circle forward is defined by either:

Input	Frames	Matching
	1-10 (or 12)	strict
	1-10	optional
	1-12	strict
	1-10 (or 12)	strict
	1-10	optional
	1-10	strict

The basic idea behind the buffering of this motion is that you always have 12f between allowed gaps (to or to) and 10f between motions that are fluid.

Some examples (assuming timing is within constraints):

-
-
-

This specific input was valid in SFV but is not anymore:

-

Otherwise, the buffers are tighter as well with 10 frames of buffering while it was 12 before (aside the last input).

Half Circle

An half circle backward is defined by either (no HCF in SF6 as far as I know).

Input	Frames	Matching
	0-12	optional*, strict
	0-12	optional*, strict
	0-12	optional*, strict
	0-12	optional*, strict
	1-12	strict

(*) You need at least **3 optional inputs** to exist.

The timing has not been validated for everyone. For instance, the range on Jamie's tenshin is 1-8. This is Blanka's roll.

Valid inputs:

-
-
-

-
-

These specific inputs were valid in SFV but are not anymore:

-
-
-

In SFV the buffer during the motion was shorter to compensate on the lenience of the motion (8 frames).

Charges

Obviously charges are never the same, here are some examples:

- Kikoken 50 frames (same)
- Spin Bird: 30 frames (40 in SFV)
- Somersault: 45 frames (same)
- Sonic Boom: 45 frames (same)
- Sumo Headbutt: 40 frames (same)

Please consult FAT Online to have details on the charge of your specific move.

Charged Super

A charged super is defined as:

Input	Frames	Matching
	50+	lenient
	1-10	lenient
	1-10	lenient
	1-4	strict

(Measured on Guile's SA)

Valid inputs

- (50)

Invalid inputs

- (50)

This is similar to SFV, but the buffer is faster now (12 frames motion, 10 frames before button).

360 / 720

Ok... I don't feel explaining it to you, right now. It seems similar to SFV.

You have to make a 360 motion in less than 21 frames. To do so, it's a combination of lenient directions and roughly covering more than an half circle.

Examples of working inputs:

-
-

-

These inputs don't work

-

I'm pretty sure the Zangiefs of the world know some weird hacks around it but everytime I look into it, it's weird.

Compared to SFV it's more strict where the whole window was 25 frames before, now it's 21 frames. **However**, compared to SFV, the motion is buffered for 20 frames whereas in SFV there is no additional buffering, the 25 frames are all you have to make the motion.

The 720 is by definition two consecutive 360.

Negative Edge

Nothing much to say but it might be useful if you don't know what it is.

It allows to trigger a motion input based on a positive edge (meaning pressing on a button), or during a negative edge (releasing a button).

It is disabled by default in SF6 and you can enable it. In SFV, it was enabled.

Example:

Do the following motion and press LP during only.

- If it happens during a buffer: it will do a fireball on reversal
- If it happens at idle: cr. LP will be thrown because there is no way of converting a cr. LP into a fireball

Kara Cancel

The Kara system is a way to promote a Fireball into a Fireball Overdrive (or EX). You have only 1 frame to input the two buttons. Similar to SFV.

Priority Resolution

Motion Matching

This is the part that looks at the input history and decides if a motion is valid or not. While we know the purest definition of each motion, in practice there are a lot of noise with other inputs in between.

To better understand, let's try to find a QCF inside: (1) (6) (1) (6) (1) (10)

In SFV, it starts from the and looks back in time:

- it sees (10) which is valid (max 12 frames)
- then it sees (1) which is also valid
- then it will look for the but there is (6) (1) (6) in between which is 13 frames
- So, **SFV doesn't see a QCF here**.

In SF6, conversely, it will try to look for all combinations in the possible frames to find *anything* that fits the definition:

- it tries to match the same inputs as SFV but it doesn't work
- it tries again by ignoring the last which produces this string of inputs (1) (6) (1) (6) (1) (10)
- after the it sees which is valid (1f+6f+1f is less than the max 11 frames)
- it sees which is valid (1f+6f is less than the max 11 frames)
- Therefore, **SF6 sees a QCF here**.

Let's look at another example of trying to find a QCF in this list of inputs. Remember that you match the inputs from top to bottom.

Input	Frames	Best match on SFV	Best match on SF6
	3	3f ≤ 12f	discarded
	3	3f ≤ 12f	discarded
	4		3f+3f+4f ≤ 11f
	8		
	1	too late: 4f+8f+1f > 12f	
	1		1f+8f+1f ≤ 11f
	1		1f ≤ 11f

This is especially important in larger motions like HCF or double QCF for supers where you have a lot of inputs that can match far back in time.

Additional note: if you have other motions attached to the the final decision might not match. In this case, this is also a valid DP motion in both SFV and SF6.

Moves Priority

Assuming that two inputs are valid at the same time, here are the main rules that are applied.

Version	Priority
SF6	Taunt > OD > SA > drive impact > throw > drive parry > 360 > DP > QC > HC > command normals > normals
SFV	Taunt > Throw > VShift > VT > VSkill > CA > 360 > (ex) HC > (ex) DP > (ex) QC > command normals > normals

You might have some local variation of those priority orders. I haven't checked all combinations but those are usually set in stone in designs and exceptions might break stuff.

Motion Conflict Resolution

Assuming Ryu with the conflict resolution between DP and QCF. This shows the difference on how the games resolve their input system. Usually it means that they consider the motion complete at different times and put a different priority.

Motion	SFV	SF6
	DP	DP
	DP	**QCF** using HC correction
	DP	**QCF** using HC correction
	DP	**QCF** using HC correction
	DP	DP
	DP	DP
	DP	DP
	DP	DP

Input Polling

XInput is read 3 times per frame with an inconsistent timing between calls (anything between 0 and 8ms typically). Assuming that you send an input within those 3 polls it is (mostly) guaranteed to be used for a frame.

In SFV or SF4 or others, it was only once per frame. Potentially, it has no major impacts, it might also be beneficial for input lag to have multiple polling routings.

> note for myself only: with an unpatched injection [1, LEFT], [1, DOWN], [1, UP] is a random between (LEFT, DOWN, UP)
> I had rare cases of missed inputs but it looks more like injection problems on my side

Summary of the main changes

	SFV	SF6
Input Buffering	Buffering Dash 8f	Buffering Dash 5f
Dash	Same	-
Down Down	12 frames buffering	10 frames buffering
Quarter Circle	12 frames buffering	11 frames buffering, can use HC inputs in some cases
Dragon Punch	8 frames buffering	7 frames buffering
Double Quarter Circle	12 frames buffering on the motion	~10 frames buffering. One diagonal is now mandatory
Half Circle	8 frames buffering on the motion but more lenient	12 frames buffering. One diagonal is now mandatory
Charges	Similar	-
Charged Super	12 motion buffer, 10 before button	10 motion buffer, 4 before button
360	25 frames for motion	21 frames for motion, same algorithm but additional 20 frames buffer for the button
Negative Edge	Always On	Off by Default, can be enabled
Kara	Same	-
Motion Matching	Backwards search	Exhaustive search
Moves Priority	Supers are lower than V-System and throws, EX are similar to Specials	OD have the highest prio and supers right after
Motion Conflict Resolution	Not the same algorithm	which leads to different outcomes